

DEEPAK GAUTAM

AI Engineer

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Education

National Institute of Technology, Kurukshetra

Bachelor of Technology in Computer Engineering (CGPA: 8.14/10)

Kurukshetra, India

Dec. 2021 – May 2025

Experience

Americana Restaurants

Data Engineer — Loyalty and Digital Systems

Mohali, India

July 2025 – Present

Peet's Coffee Loyalty Platform | C#, Azure Functions, Zoho Creator, REST APIs

- Architected a backend rewrite in **C#** to migrate a legacy reward model to a scalable **3-tier loyalty system** for **50,000+ daily active users**.
- Engineered an **earn–burn loyalty engine** using tier-based multipliers and dynamic redemption logic to process real-time checkout discounts.
- Validated cross-platform data integrity across **Azure and Zoho Creator**, preventing transaction discrepancies across distributed systems.

Americana Restaurants

Data Engineer Intern / AI Developer – Generative AI

Mohali, India

Jan. 2025 – June 2025

Sales Deep Analyzer (RAG Pipeline) | GPT-4o, Azure AI Search, Python, LangChain

- Built an **NLP-to-SQL analytics assistant** for leadership to query sales trends across **MENA brands (KFC, Hardee's, Pizza Hut)** via natural language.
- Architected a **RAG pipeline** using **GPT-4o and Azure AI Search** to translate business queries into optimized SQL datasets from **SQL Server**.
- Developed a **self-service UI** that auto-generates visual dashboards from query outputs with support for Excel exports and user feedback.

Technical Skills

Languages: C, C++, Python, C#, JavaScript, TypeScript, SQL

Frameworks & Libraries: React, Next.js, Node.js, FastAPI, TensorFlow, Keras, Pandas, LangChain

Databases & Cloud: PostgreSQL, Supabase, pgvector, PostGIS, SQL Server, SQLite, Azure (Functions, AI Search), Docker, Git

AI & Systems: RAG, Agentic AI, MCP/FastMCP, Groq, SSE Streaming, NLP-to-SQL, Prompt Engineering, Vector Search, REST APIs

Projects

SafarSquad — Social Travel & AI Platform | React, TypeScript, FastAPI, Supabase, Groq Source Code | Live Demo

- Engineered a production-grade **social travel PWA** enabling users to discover trips via interactive **Leaflet maps**, manage join requests, and build communities, complemented by a **referral-based rewards system**.
- Developed a highly interactive user experience featuring **real-time group chat** and live push notifications. Integrated post-trip features including star-rating reviews and secure, participant-only **photo galleries**.
- Architected an **agentic AI travel assistant** (Llama-3.3-70B) with **voice-to-text** input and **page-context injection**, autonomously triggering a **FastMCP weather server** to deliver live forecasts via sub-second **SSE** streams.
- Built a **hybrid search engine** combining **pgvector** semantic similarity with **PostGIS** spatial filtering, with **Webhooks** automating real-time vector indexing; secured all data via **OAuth** and **Row-Level Security (RLS)**.

JobPilot – Automated Job Hunting | Python, MCP Protocol, Claude AI, JobSpy, SQLite

Source Code

- Built an **automated job-hunting pipeline** that scrapes 7+ platforms (Indeed, LinkedIn, Amazon) nightly with zero manual intervention.
- Engineered a **resume-aware scoring engine** that ranks jobs using regex matching and filters out senior roles via experience-based word boundaries.
- Developed **custom MCP servers** enabling Claude AI to accept natural language queries over live Gmail data via SQLite database integration.

Honors and Achievements

- **Amazon ML Summer School 2024:** Selected from over **90,000+ applicants** for an elite Machine Learning mentorship program led by Amazon scientists.
- **Elite Competitive Programming Ranks:** Achieved a peak **Global Rank of 31** on CodeChef and **Global Rank 201** in a worldwide Codeforces contest.
- **Platform Ratings:** **Knight** on LeetCode (Top 5%), **Expert** on Codeforces (1665), and **4-Star** on CodeChef. Solved **2,500+ DSA problems**.